LETTERS
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Estimated carbon dioxide emissions from tropical deforestation improved by carbon-density maps

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Deforestation contributes 6–17% of global anthropogenic CO_2 emissions to the atmosphere[1]. Large uncertainties in emission estimates arise from inadequate data on the carbon density of forests[2] and the regional rates of deforestation. Consequently there is an urgent need for improved data sets that characterize the global distribution of aboveground biomass, especially in the tropics. Here we use multi-sensor satellite data to estimate aboveground live woody vegetation carbon density for pan-tropical ecosystems with unprecedented accuracy and spatial resolution. Results indicate that the total amount of carbon held in tropical woody vegetation is 228.7 Pg C, which is 21% higher than the amount reported in the *Global Forest Resources Assessment 2010* (ref. 3). At the national level, Brazil and Indonesia contain 35% of the total carbon stored in tropical forests and produce the largest emissions from forest loss. Combining estimates of aboveground carbon stocks with regional deforestation rates[4] we estimate the total net emission of carbon from tropical deforestation and land use to be 1.0 Pg C yr^{-1} over the period 2000–2010—based on the carbon bookkeeping model. These new data sets of aboveground carbon stocks will enable tropical nations to meet their emissions reporting requirements (that is, United Nations Framework Convention on Climate Change Tier 3) with greater accuracy.

When forests are cleared, carbon stored above and below ground in leaves, branches, stems and roots is released to the atmosphere. As a consequence, forest clearing, especially in the tropics, is a major source of CO_2 to the atmosphere. Although the proportion of carbon stored in forests comprises 70–80% of total terrestrial carbon[5], the spatial and temporal variability in carbon storage is substantial[6]. This variability arises from natural and anthropogenic disturbances, as well as differences in stand age, topography, soils and climate. Globally, soils hold two to three times more carbon than that stored above ground in forest vegetation, but with the exception of cultivation, peatland fires and thawing permafrost, much of the carbon in soils is physically and chemically protected and not easily oxidized[7]. In contrast, carbon stored in aboveground biomass is readily mobilized by disturbance processes such as fire, wind throw, pest outbreaks and land conversion[8].

Efforts to quantify the amount of carbon stored in aboveground biomass over large areas of the tropics have been fraught with uncertainty. For example, estimates of aboveground carbon storage in tropical African forests vary by over 100% (46.9 Pg–104.5 Pg; ref. 9). In turn, the lack of reliable estimates of forest carbon storage introduces large uncertainties into estimates of terrestrial carbon emissions[10–14]. In Amazonia, recent studies have suggested that as much as 60% of the total uncertainty in estimated emissions can be attributed to uncertainty in the carbon stocks for deforested lands[15].

The *Global Forest Resources Assessment* (FRA) published by the United Nations Food and Agriculture Organization (FAO) provides a comprehensive accounting of aboveground carbon stocks for tropical forests and other wooded lands. The most recent national estimates, reported in the FRA 2010 (ref. 3), are derived primarily from ground-based forest inventories. However, the majority of existing forest inventories are not current. Furthermore, because these inventories were designed to provide estimates of commercial timber volume, the tree species included and the measurements acquired reflect a bias towards the fraction of aboveground biomass that is merchantable. Moreover, many countries do not undertake forest inventories or their inventories are obsolete. To reduce the uncertainty in estimates of carbon emissions resulting from deforestation and forest degradation, more complete and higher quality information on the spatial distribution of carbon stocks is needed.

Here we estimated the carbon density (Mg C ha^{-1}) of aboveground live woody vegetation for the pan-tropics (including tropical Africa, America and Asia) at a spatial resolution of 500 m using a combination of remote sensing and field data (for the period 2007–2008; Fig. 1). Specifically, we used satellite-based light detection and ranging (LiDAR) data acquired at a nominal spatial resolution of 70 m together with 500 m multispectral surface reflectance imagery and other geospatial data layers (see Methods). Using a multi-scale calibration and mapping strategy, we produced the first wall-to-wall map of aboveground biomass at 500 m resolution for the pan-tropics. Field measurements were collected from forests across tropical Africa, America and Asia from 2008 to 2010 at sample points co-located with LiDAR 'footprints' using a sampling protocol specifically designed for the optimal integration of field and satellite data. LiDAR waveform measurements were acquired by the Geoscience Laser Altimeter System (GLAS) onboard NASA's Ice, Cloud, and land Elevation Satellite (ICESat; ref. 16). Surface reflectance data were provided by the Moderate Resolution Imaging Spectroradiometer (MODIS) onboard NASA's Terra and Aqua satellites and digital elevation data were obtained as part of NASA's Shuttle Radar Topography Mission (SRTM). The methods used to produce the data set shown in Fig. 1 are provided in the Methods section and in the Supplementary Information.

We estimate the total mass of carbon stored above ground in live woody vegetation of tropical America, Africa and Asia to be 117.7 (±8.4), 64.5 (±8.4) and 46.5 (±3.0) Pg C, respectively, at

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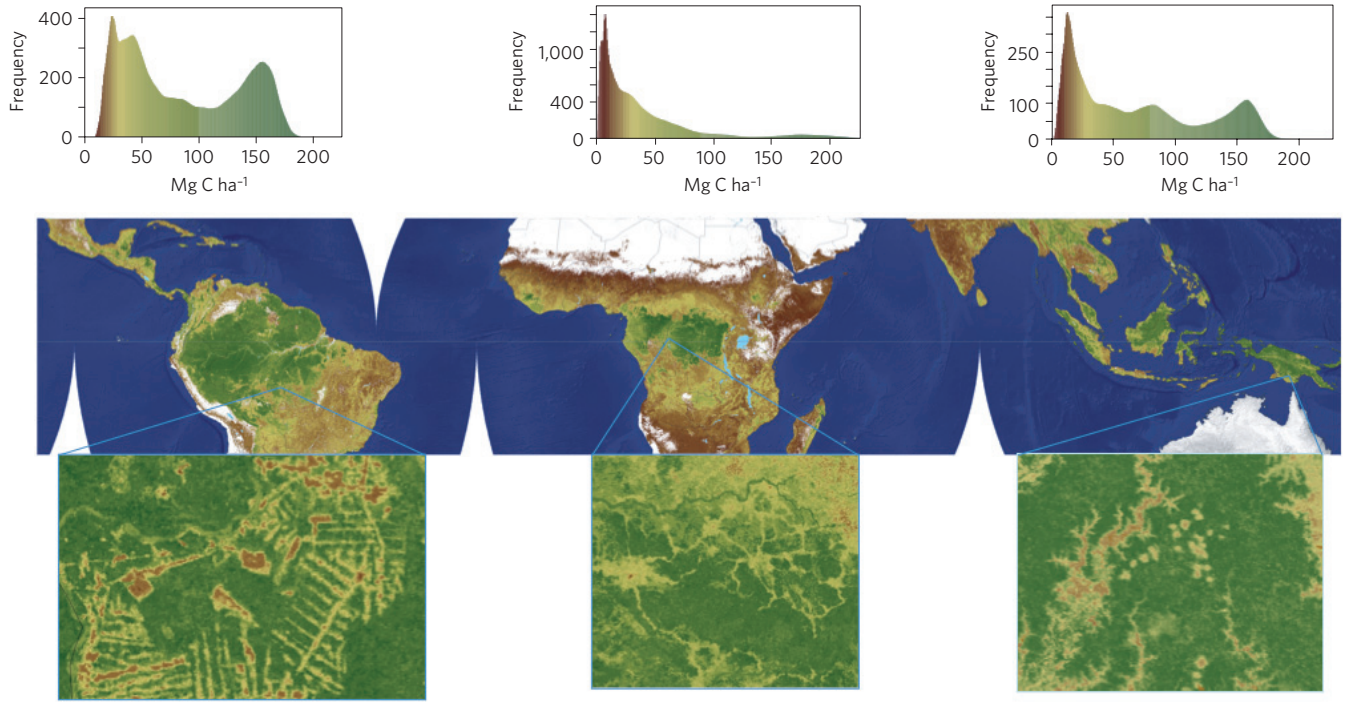


Figure 1 | Carbon contained in the aboveground live woody vegetation of tropical America, Africa and Asia (Australia excluded). The upper panels show the frequency distribution of carbon in units of Mg C ha^{-1} for each region. Inset figures across the bottom provide higher-resolution examples of the spatial detail present in the satellite-derived biomass data set. Carbon amount is represented in the maps as a colour scheme from dark brown (low carbon) to dark green (high carbon). See upper panels for numeric values.

a 95% confidence interval (Supplementary Section S3.1). These estimates are 21% higher than estimates provided by the FRA 2010, the most comprehensive data set of tropical forest carbon compiled to date. At the national level (Supplementary Appendix S1), we estimate that the forests of Brazil and Indonesia store 53.2 and 18.6 Pg C, respectively, despite high historical deforestation rates[4]. The Democratic Republic of the Congo, which has experienced relatively low rates of forest loss, is second only to Brazil in having the largest stock of carbon (22.0 Pg C) in forests. When our results are compared with estimates derived from more recent and redesigned national forest inventories implemented by the FAO for a limited number of tropical countries[17], (Fig. 2; Supplementary Section 3.2), there is close agreement, providing further support for the conclusion that carbon storage in tropical forests is substantially greater than previously thought. We attribute this result to the ability of our approach to capture the range and geographic distribution of carbon density across the tropics with a degree of accuracy that was not previously feasible (Supplementary Fig. S13). This accuracy varies with the spatial scale of analysis (Supplementary Sections 3.1, 3.2) and we expect accuracies to increase in the decade ahead as new satellite sensors (for example, satellite LiDAR, and radar) designed specifically for vegetation structure measurements become available for ecosystem studies.

The spatially explicit nature of the new carbon-density data set allows comparisons to be made among regions and nations that were not possible in the past, and also provides important insights into how aboveground carbon is distributed across land cover types both locally and regionally. Tropical forests store 84.2% of aboveground carbon in Latin America, while shrublands and savannahs (for example, Brazilian Cerrado) store 15.8% (Table 1). Aboveground carbon is partitioned similarly in Southeast Asia, but in Africa carbon storage is more evenly distributed between forests (54.1%) and other woody vegetation (45.9%; consistent with ref. 18). As a result, the clearing of non-forested lands in Africa may contribute significantly to total carbon emissions.

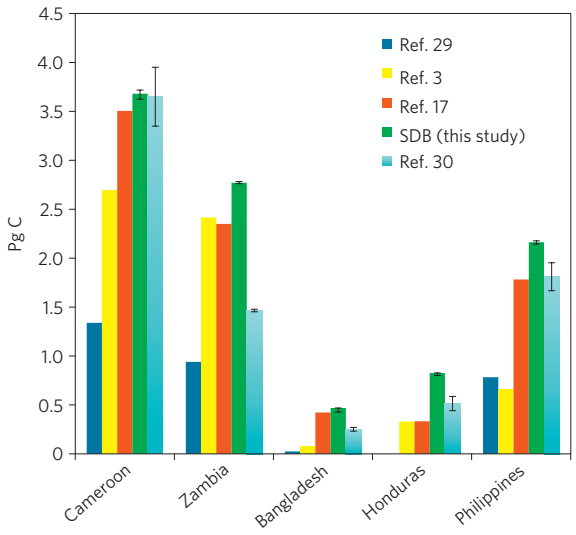


Figure 2 | Comparison of national aboveground carbon stock estimates. The figure shows five tropical nations for which FAO FRA 2005 (ref. 29), FAO FRA 2010 (ref. 3), FAO National Forest Monitoring and Assessment (NFMA) (ref. 17) and SDB estimates are available. Alternative carbon stock estimates[30] are shown for comparison. The error bars indicate the uncertainty in national level estimates (at 95% CI for SDB data).

In addition to providing information on the geographic distribution of aboveground carbon stocks, the new data provide an improved basis for estimating CO_2 emissions from tropical deforestation. Towards this end, we used a well-established model[19,20] to calculate pan-tropical carbon emissions for the period 2000–2010. Three estimates of aboveground live biomass (that is, carbon density) served as alternative inputs: satellite-derived biomass (SDB; Fig. 1), satellite-derived biomass weighted

Table 1 | Total aboveground carbon stocks in Pg C (and per cent of total) across tropical Africa, America and Asia.

	Tropical Africa	Tropical America	Tropical Asia
Forest	32.4 (54.1)	91.2 (84.2)	35.4 (85.3)
Shrubland and savannahs	27.5 (45.9)	17.1 (15.8)	6.1 (14.7)
Total	59.8 (100)	108.3 (100)	41.5 (100)

Forest (International Geosphere–Biosphere Programme (IGBP) classes 1–5) and other woody vegetation (IGBP classes 6–9).

by deforestation (SDBdw; Supplementary Section S4) and FRA 2010 biomass[3]. The resulting estimates of net CO_2 emissions were then compared with earlier results from[21,22] (Fig. 3, Table 2).

The new SDB data set yielded estimates of total net CO_2 flux lower than recent estimates[21–23] but higher than estimates based on FRA 2010 (Fig. 3, Table 2). Relative differences among modelled fluxes were generally less than differences in carbon densities, partly because emissions of CO_2 from soil cultivation were held constant throughout the simulations and were thus unaffected by differences in aboveground carbon density. The difference observed between emissions based on refs 21 and 22 arise from different rates of deforestation (lower rates reported in the FAO FRA 2010; ref. 3). Differences between SDBdw, ref. 21 and FAO FRA 2010 (ref. 3) result from different estimates of mean carbon density.

It is important to note that the net emissions calculated with the bookkeeping model include more than deforestation. They include the gross uptake of carbon in secondary forests (recovering from wood harvest; 0.48 Pg C yr^{-1}) and in the fallow cycle of shifting cultivation (0.71 Pg C yr^{-1}), as well as the gross emissions of carbon from burning (0.78 Pg C yr^{-1}), from harvested wood products (0.46 Pg C yr^{-1}), from dead plant material left on site after deforestation (0.83 Pg C yr^{-1}) and from cultivated soils (0.15 Pg C yr^{-1}). The estimated gross and net emissions of carbon from deforestation are the same (1.14 Pg C yr^{-1}), whereas the gross emissions from all uses of land (2.22 Pg C yr^{-1}) are more than two times larger than the net flux. These gross emissions are largely offset by the gross sinks in secondary forests (recovering from wood harvest) and in the fallow cycle of shifting cultivation.

The spatial information afforded by the satellite-derived carbon-density map presented here, in combination with the improved estimate of total average carbon density, provides a strong foundation for estimating fluxes relative to previously available data sets[24,25]. The emissions estimate represents an improvement because (1) the estimates of biomass density for each region are derived from field calibrated LiDAR combined with wall-to-wall satellite measurements rather than from sparse field plots and (2)

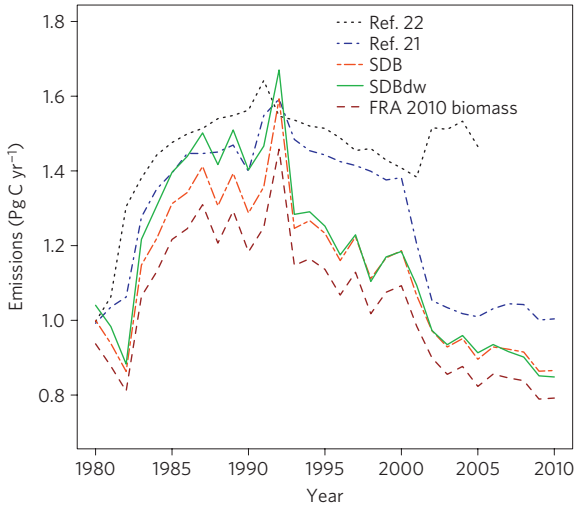


Figure 3 | Annual net emissions of carbon from land-use change in the tropics. The average carbon densities given by ref. 21, SDB, SDBdw and FRA 2010 (ref. 3) differ. An earlier estimate from ref. 22 is shown for comparison. The rates of deforestation given by refs 22 and 21 differ (the former is based on FRA 2005 (ref. 29) and the latter on FRA 2010 (ref. 3).

the estimates of biomass density were further refined to reflect the biomass density of those areas actually deforested, as derived from the co-location of carbon-density and deforestation maps[4]. Specifically, the spatially resolved nature of the carbon-density data set allowed the calculation of an area-averaged flux weighted by the area undergoing deforestation (SDBdw; Supplementary Section S4), rather than by forest area alone (SDB). In tropical Asia, for example, deforestation took place in areas with lower than average carbon density (SDB versus SDBdw), whereas in tropical America the converse was true: deforestation occurred in forests where average aboveground carbon density was higher than the regional average. In Africa the small net difference for the entire region was, again, the result of offsetting effects among subregions (Supplementary Fig. S18).

Deforestation in tropical America is widely reported to have occurred in lands with lower than average carbon density[26]. Our results show that this finding is dependent on the ecosystems that are included in the estimation of the average density. Specifically, the arc of deforestation in the Brazilian Amazon is located in areas dominated by transitional forests of lower biomass than many Amazonian forests[26]. However, the average carbon density for woody vegetation reported here (SDB) for tropical America includes not only dense forest vegetation but also a range of transitional forest areas, including widespread areas of cerrado (that is, woody savannah) vegetation. As a consequence, transitional forests on the edge of Amazonia have an average carbon density that

Table 2 | Estimates of carbon density (Mg C ha^{-1}) and flux (Pg C yr^{-1}) for the period 2000–2010 resulting from alternative sources of carbon-density information.

	Tropical Africa		Tropical America		Tropical Asia		Total Tropics	
	Stock	Flux	Stock	Flux	Stock	Flux	Stock	Flux
Ref. 21	66	0.31	134	0.49	160	0.27	117	1.07
SDB	82	0.30	116	0.47	119	0.18	105	0.95
SDBdw	92	0.27	137	0.56	66	0.13	109	0.96
FRA 2010 Biomass	87	0.29	104	0.43	93	0.16	96	0.88
Ref. 22	66	0.25	134	0.63	160	0.60	117	1.47

SDB reflects carbon density from Fig. 1 aggregated regionally. SDB was also weighted by areas where deforestation took place (SDBdw).

is higher than many other areas dominated by woody vegetation across tropical South America.

These results suggest that the recent estimate of 1.1 Pg C yr^{-1} released from tropical deforestation during the period 2000–2010 (ref. 21) overestimates the net flux of carbon by 11–12% (Table 2). Flux models capable of more fully using information on the spatial distribution of deforestation and carbon density will allow further refinement of carbon emission estimates from tropical nations.

Methods

To characterize the magnitude and spatial distribution of aboveground carbon contained in live woody vegetation we used field data to calibrate spaceborne LiDAR observations which were then used to generate a spatially continuous data set of carbon storage at a nominal spatial resolution of 500 m. We employed a three-stage data collection and modelling strategy which was specifically designed to allow measurements collected on the ground to be up-scaled to the resolution of the MODIS 500 m imagery using the GLAS LiDAR data. In the first stage, we use a field-tested measurement protocol to characterize aboveground woody live biomass within GLAS footprints (nominally 70 m in diameter) over a broad range of conditions in tropical Africa, America and Asia. A protocol was established to standardize data collection from field campaigns on three continents, with an emphasis on obtaining measurements of stem diameter (diameter at breast height; DBH) of all live trees having a DBH ≥ 5 cm occurring within, and centred on, individual GLAS footprints.

In the second stage, allometric relationships were used that allowed for the estimation of tree biomass density from the field-measured data (after ref. 27; Supplementary Section S1.1). Tree biomass density was converted to carbon using a coefficient of 0.5 (ref. 27). A statistical relationship between the field biomass estimates and LiDAR waveform metrics was then established, effectively allowing for the extension of the field measurements to thousands of new, unsampled locations (that is, GLAS footprints) across the tropics (Supplementary Section S1.2).

In the third stage we use thousands of GLAS-based estimates of biomass to calibrate the statistical models that are used to generate the wall-to-wall, pan-tropical biomass map from a best-quality temporal composite of MODIS Nadir BRDF (Bidirectional Reflectance Distribution Function)-Adjusted Reflectance (NBAR), MODIS Land Surface Temperature (LST) temperature and SRTM digital elevation data using the Random Forest machine learning algorithm[28]. Error estimates were based on a 10% independent sample of GLAS-based biomass estimates reserved for each continent (Supplementary Fig. S14).

Carbon fluxes were calculated using the same bookkeeping model employed in previous analyses[19–23]. The flux model tracks annual changes in carbon density when forest area is cleared for cropland, pasture and shifting cultivation, when forests are harvested, when plantations are established, and when agricultural lands are abandoned (and returned to forest). The sum of all changes in carbon for all areas under management defines the annual net flux of carbon from land-use change. To calculate the net emissions from land-use change, the non-spatial bookkeeping model was used together with rates of deforestation from FAO (2010) and biomass density for each region obtained by weighting biomass density by the areas deforested (based on ref. 4). If the biomass density of the forests that have been removed can be calculated, as was accomplished here, then the emissions can be estimated non-spatially. An advantage of the bookkeeping model, as well as associated data on wood harvest (degradation) and secondary forest growth (recovering from harvest and shifting cultivation), is that many more of the processes that determine the net flux of carbon from land use can be accounted for rather than simply gross deforestation.

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Author contributions

A.B., N.T.L., W.S.W., S.J.G. and R.A.H. designed the study. A.B., M.S., J.H. and D.S-M. conducted the analysis. A.B., R.D., S.S. and P.S.A.B. designed and conducted the error analysis. A.B., S.J.G., R.A.H., W.S.W. and M.A.F. wrote the paper.

Additional information

The authors declare no competing financial interests. Supplementary information accompanies this paper on www.nature.com/natureclimatechange. Reprints and permissions information is available online at http://www.nature.com/reprints. Correspondence and requests for materials should be addressed to A.B.